

Streaming Communication for Scalable Data Exchange in DYNAMOS

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Abstract—This thesis investigates how streaming communication can be introduced into DYNAMOS, a policy-driven middleware for data exchange based on dynamically composed microservices. The current system relies on unary gRPC request-response interactions, which become increasingly inefficient for large or continuous transfers. To ground the broader architectural and comparative study, the thesis first reviews candidate streaming mechanisms and evaluates them in a controlled *Producer* → *Transformer* → *Sink* pipeline. The headline empirical study is a single-node Kubernetes benchmark comparing client-streaming gRPC, unary gRPC, RabbitMQ Streams, NATS JetStream, and Kafka across a clean bulk-transfer matrix and three diagnostic scenarios (continuous steady-rate flow, slow-consumer backpressure, and forced-restart recovery). An exploratory three-broker clustered deployment, in which each broker type runs as a replicated StatefulSet across three worker nodes, is reported as a supporting observation rather than as headline evidence. In the single-node benchmark, client-streaming gRPC provides the highest sustained throughput on the clean matrix and the most stable pacing in the continuous scenario. Among the brokered transports, NATS JetStream achieves the highest clean-matrix throughput average once JetStream resource limits, async publish, and batched pull consumption are configured correctly; RabbitMQ Streams offers stable throughput under clean conditions but accumulates large tail latency under continuous and backpressure load; Kafka delivers duplicate-free and ordering-violation-free recovery, at the cost of heavier tail latency and higher broker resource usage. The clustered runs suggest that brokered transports benefit measurably from horizontal deployment, particularly Kafka and RabbitMQ under pressure, but coverage is limited and the headline conclusions are drawn from the single-node evidence. These findings establish a concrete trade-off map across throughput, latency, delivery semantics, and recovery behaviour for the purpose of selecting a streaming communication model for DYNAMOS.

Index Terms—streaming communication, microservices, gRPC, event streaming, policy-driven middleware, DYNAMOS

I. INTRODUCTION

Many contemporary distributed systems operate in environments where data and computational resources are owned by different organizations. In such settings, data exchange must adhere to legal agreements, privacy regulations, and organizational policies. These constraints are common in domains such as healthcare, scientific research, and public-sector infrastructures, where unrestricted data exchange is not acceptable.

DYNAMOS (Dynamically Adaptive Microservice-based Operating System) is a data exchange middleware developed by the Complex Cyber-Infrastructure (CCI) group at the

University of Amsterdam. The system implements a set of recurring data sharing archetypes that describe how data and computation are distributed among participants under policy constraints. DYNAMOS realizes these archetypes by dynamically composing chains of microservices that execute data sharing requests in compliance with formally evaluated policies. The middleware is designed to be adaptive, allowing architectural and execution decisions to change at runtime in response to policy requirements.

The current implementation of DYNAMOS uses gRPC for inter-service communication and relies exclusively on atomic request-response interactions. This communication model is suitable for small or bounded data transfers but becomes limiting when applied to large-scale or continuous data exchange. In practice, data in distributed systems is often produced and consumed incrementally, for example in sensor data streams, distributed analytics pipelines, or staged processing of large research datasets.

When request-response communication is used in such scenarios, systems must either buffer large volumes of data before transmission or split transfers into multiple independent requests. Buffering increases memory usage and can lead to idle time between producers and consumers. Repeated request-response interactions introduce additional coordination overhead. Both effects negatively impact scalability and resource utilization as data volumes and the number of participating services increase.

Streaming-based communication has been proposed as a solution to these limitations. Streaming allows data to be transferred and processed incrementally, enabling overlap between computation and communication and reducing peak memory usage. gRPC supports several streaming interaction patterns, including client-side, server-side, and bidirectional streaming.

Integrating streaming communication into DYNAMOS is not straightforward. Streaming relies on long-lived connections and continuous data flows, which must be reconciled with DYNAMOS' policy-driven execution model, dynamic microservice composition, orchestration decisions, and fault handling mechanisms. Without careful design, the introduction of streaming may interfere with policy enforcement or reduce the ability to evaluate and reason about system behavior.

The goal of this thesis is to investigate how streaming communication can be incorporated into DYNAMOS in a way that improves scalability and performance while preserving

policy compliance and adaptive behavior. This involves analyzing existing streaming communication mechanisms, designing architectural extensions to the middleware, implementing streaming support, and evaluating the resulting system under realistic workloads.

The main research question addressed in this thesis is:

Main RQ: How can streaming communication be integrated into a dynamically adaptive, policy-driven microservice-based data exchange middleware such as DYNAMOS, while preserving compliance guarantees and improving scalability and performance?

To answer this question, the following sub-research questions are defined:

- **RQ1:** Which streaming communication technologies, frameworks, and interaction patterns are suitable for large-scale and continuous data exchange in microservice-based systems such as DYNAMOS?
- **RQ2:** How can streaming communication be integrated into a microservice-based architecture in a way that preserves dynamic service composition, policy-driven orchestration, and isolation guarantees?
- **RQ3:** What is the impact of streaming communication on scalability, performance, and robustness compared to a traditional request-response approach under realistic workloads in a microservice-based architecture?

II. RELATED WORK

This section reviews existing literature related to streaming communication in microservice-based systems, with a particular focus on topics that are central to this thesis: streaming communication technologies and interaction models, architectural integration in adaptive microservice middleware, policy enforcement under continuous data exchange, resilience mechanisms for streaming workflows, and validation practices with respect to scalability and performance.

A. Streaming communication technologies and interaction models

DYNAMOS is implemented as a dynamically composed microservice system deployed on Kubernetes, using gRPC for inter-service communication and RabbitMQ for asynchronous messaging [1], [2]. This constrains the design space for introducing streaming functionality to a limited set of technologies that are compatible with containerized microservices and cloud-native deployment models. The primary question addressed in this body of work is which streaming communication mechanisms are suitable for large-scale or continuous data exchange, and under which conditions direct streaming RPC should be preferred over broker-based approaches.

gRPC provides native support for streaming through client-streaming, server-streaming, and bidirectional streaming RPCs. These interaction patterns differ in message flow control, lifetime of connections, buffering behaviour, and error propagation semantics. In the context of DYNAMOS, the choice of streaming pattern is not merely an API concern but has architectural consequences, as it affects memory usage,

cancellation behaviour, and the ability to overlap communication and computation. The original DYNAMOS thesis identifies client-streaming as a promising mechanism for transferring large datasets incrementally, reducing peak memory pressure at the cost of increased handling complexity [1].

However, the use of long-lived streaming connections introduces operational challenges that are largely absent in unary request-response interactions. Starck and Ghofrani show that gRPC streaming complicates load balancing in Kubernetes environments, as persistent connections can cause traffic to become unevenly distributed across replicas [3]. This observation is particularly relevant for systems such as DYNAMOS that rely on horizontal scaling and ephemeral execution units. Complementary benchmarking work demonstrates that the performance characteristics of streaming gRPC are sensitive to runtime configuration and virtualization overhead, with measurable effects on latency and resource utilisation in containerized environments [4].

An alternative to direct service-to-service streaming is to externalise the data plane to a messaging or event streaming platform. Broker-based communication decouples producers and consumers in time, delegates buffering and persistence to the broker, and can simplify fan-out scenarios. Comparative benchmarks of Kafka, RabbitMQ, NATS, and ActiveMQ Artemis show that different brokers dominate under different workload characteristics, reinforcing that broker choice is highly workload dependent [5]. A cloud-native literature review comparing Kafka, Pulsar, and RabbitMQ further highlights trade-offs between throughput, reliability, operational complexity, and Kubernetes integration [6]. While these studies are not conducted in the context of policy-driven middleware, they provide structured criteria that are directly applicable when evaluating broker-based alternatives for DYNAMOS.

Several studies also emphasize that streaming design is influenced not only by technology choice but by the underlying communication model. DataXc contrasts push-based streaming approaches with pull-based designs and reports improved latency stability and throughput under consumer-controlled flow [7]. Although the application domain differs, the distinction between producer-driven and consumer-driven streaming is directly relevant to backpressure management and resource utilisation in DYNAMOS-compatible workflows.

Overall, existing literature provides a broad comparison of streaming technologies and interaction patterns, but these comparisons are typically conducted in generic microservice or stream-processing settings. There is limited work that evaluates these mechanisms under the specific constraints of dynamically composed, policy-driven middleware with ephemeral execution units, leaving open questions regarding their suitability in such environments.

B. Architectural integration of streaming in adaptive microservice middleware

Beyond the selection of a streaming mechanism, integrating streaming communication into an adaptive microservice

architecture raises architectural questions related to modularity, orchestration, and lifecycle management. DYNAMOS is explicitly designed around dynamic microservice composition, where execution chains are generated and deployed per request based on policy evaluation and system state [1], [2]. These chains are short-lived by design, which improves isolation and compliance guarantees but complicates the management of long-lived communication channels.

Communication in DYNAMOS is abstracted through a sidecar-based architecture, decoupling communication concerns from core microservice logic. The sidecar pattern is widely used in cloud-native systems to address cross-cutting concerns such as observability, security, and traffic management. Figueira and Coutinho demonstrate how sidecars can be integrated into self-adaptive microservice architectures on Kubernetes, enabling runtime adaptation without violating service modularity [8]. While their work does not explicitly address streaming data transfer, it illustrates how communication behaviour can be adapted dynamically at runtime.

Several middleware frameworks operationalise sidecars as first-class runtime components. Figueira and Coutinho discuss sidecar-based runtimes such as Dapr, which provide a protocol-agnostic communication abstraction supporting service invocation, pub/sub, and state management through a sidecar process [8]. Such runtimes demonstrate that heterogeneous communication paradigms, including streaming and messaging, can be unified behind a stable interface. However, this class of runtimes assumes relatively static service topologies and does not address dynamic chain composition driven by policy evaluation, which limits their applicability as a direct solution for DYNAMOS.

Existing architectural work therefore supports the feasibility of integrating streaming into sidecar-based microservice systems but does not address the combination of long-lived streaming connections with ephemeral, dynamically composed execution chains. This limitation is particularly relevant in systems that combine adaptivity with strict policy enforcement requirements.

C. Policy enforcement and compliance under continuous data exchange

Policy enforcement is a defining characteristic of DYNAMOS, where access rights, execution locations, and data-flow constraints are evaluated prior to execution and enforced through dynamically composed microservice chains [1], [2]. In the current architecture, policy enforcement is implicitly scoped to bounded request-response interactions. Introducing streaming communication challenges this assumption, as data transfer may span longer periods during which policies, system state, or execution context may change.

Neumaier et al. propose a policy-aware architecture for decentralized dataset exchange that emphasizes continuous monitoring and runtime validation against machine-readable policies [9]. While their work does not explicitly target streaming RPC, it introduces an important distinction between one-time policy clearance and ongoing compliance monitoring,

which becomes critical when data is exchanged incrementally over long-lived connections.

At the communication layer, several studies propose enforcing security and access control through proxies or sidecars. Thiagarajan et al. present a proxy-based approach for securing gRPC communication using mTLS, JWT authentication, and RBAC [10]. Although the focus is on security rather than data usage policy, the architectural approach is relevant, as it suggests that enforcement can be applied at communication boundaries and potentially at message granularity.

Despite these contributions, there is limited guidance on how policy enforcement should interact with long-lived streaming connections in systems that dynamically compose execution chains. In particular, existing work does not address how policy violations should be detected or handled mid-stream, or how streams should be terminated or adapted when compliance conditions change. These limitations highlight open challenges when introducing streaming into policy-driven middleware such as DYNAMOS.

D. Resilience, backpressure, and fault handling in streaming workflows

Streaming-based communication introduces operational challenges that are less pronounced in request-response systems. Long-lived data flows must tolerate partial failures, adapt to fluctuating load, and prevent uncontrolled resource growth through effective backpressure mechanisms. These challenges are compounded in DYNAMOS by dynamic microservice composition and ephemeral execution units.

Reactive microservice architectures emphasize asynchronous communication, non-blocking execution, and fault isolation as mechanisms for improving resilience under variable load [11]. Although this work does not focus on streaming RPC, its principles are directly applicable to streaming workflows, where failures and backpressure must be managed continuously rather than per request.

Studies on high-volume data processing in microservices highlight the importance of explicit flow control and decoupling between producers and consumers. Guntupalli and Vamshi show that event-driven architectures combined with messaging systems can mitigate bottlenecks and stabilize performance under sustained data rates [12]. While these systems typically rely on brokers rather than direct streaming RPC, they reinforce the need for explicit backpressure mechanisms in streaming data paths.

From a systems perspective, Gan and Delimitrou demonstrate that communication overhead and tail latency dominate the performance of large-scale microservice deployments [13]. Their findings underline that resilience and scalability issues often emerge from inter-service communication rather than computation, an effect that is amplified in streaming scenarios.

Existing literature provides well-established principles for resilience and fault tolerance in microservice systems, but these are typically studied either in request-oriented architectures or broker-based streaming platforms. There is limited work examining how these mechanisms interact with direct

streaming RPC in dynamically composed microservice chains, leaving questions about how failures and backpressure should be handled in dynamically composed microservice systems.

E. Validation practices, scalability, and performance evaluation

Empirical evaluation of streaming communication often focuses on comparing streaming and non-streaming interaction styles in terms of throughput, latency, and resource usage. Studies focused specifically on gRPC streaming show that streaming RPCs outperform unary calls for large payloads and continuous data flows by enabling incremental processing and reducing buffering overhead [4], [14]. Complementary protocol-level comparisons between REST and gRPC further establish that binary, multiplexed transports outperform request-response HTTP under load, which provides additional context for choosing gRPC as the baseline communication layer [15], [16].

Scalability in containerized streaming systems has been studied through systematic mapping and empirical evaluations. A mapping study on performance analysis techniques for streaming workloads in containerized microservices identifies network contention, scheduling overhead, and coordination costs as primary scalability bottlenecks [17]. Empirical studies of streaming pipelines deployed on Kubernetes further show that streaming approaches scale more gracefully than request-response models when concurrency increases, provided that backpressure and flow control are properly configured [3].

However, most existing evaluations assume static microservice topologies and fixed execution paths. Prior performance studies rarely consider adaptive middleware that dynamically constructs execution chains based on policy evaluation. As a result, there is limited empirical insight into how streaming communication scales when embedded within dynamically adaptive, policy-driven systems such as DYNAMOS. This lack of combined evaluation of streaming, scalability, and adaptivity motivates the experimental focus of this thesis.

III. CANDIDATE MECHANISMS AND BENCHMARK DESIGN

This section addresses *RQ1*: which streaming communication technologies, frameworks, and interaction patterns are suitable for large-scale and continuous data exchange in microservice-based systems such as DYNAMOS? It combines a literature-backed comparison of candidate mechanisms with a controlled benchmark design that establishes a gRPC reference baseline before broker-based alternatives are added.

A. gRPC streaming

According to the official gRPC documentation, gRPC is based on service definitions in Protocol Buffers and supports four RPC styles: unary, server-streaming, client-streaming, and bidirectional streaming. For streaming calls, message ordering is preserved within each individual RPC, while deadlines, cancellation, metadata, and both synchronous and asynchronous APIs are supported by the runtime [18]. These properties

make gRPC streaming the most direct extension of the current DYNAMOS communication model.

For microservice chains with predominantly point-to-point data transfer, client-streaming is the most natural first candidate because it lets a sender incrementally push chunks or records to a single downstream consumer. Bidirectional streaming becomes relevant when the receiver should provide progress feedback, pacing information, or explicit acknowledgements during transfer. The main advantages of gRPC streaming are low protocol overhead, direct reuse of protobuf contracts, and the absence of an additional broker layer. The disadvantages are that sender and receiver remain temporally coupled, durable replay is not provided by default, and backlog handling or crash recovery must be implemented at application level. Even so, prior work on asynchronous gRPC streaming demonstrates that carefully designed protocols can support efficient and even exactly-once style delivery patterns, which makes gRPC a strong experimental baseline rather than merely the status-quo option [4], [14], [15].

B. Apache Kafka

Apache Kafka is officially described as an open-source distributed event-streaming platform for high-performance pipelines, streaming analytics, data integration, and mission-critical applications. Its published core capabilities emphasise high throughput, durable storage, high availability, built-in stream processing, and large-scale horizontal scalability [19]. In architectural terms, Kafka exposes an append-only log abstraction in which messages are written to topics and partitions, allowing replay and independent consumption by multiple downstream readers [20].

Kafka is therefore the strongest candidate when temporal decoupling, replayability, or large consumer backlogs matter more than direct point-to-point simplicity. Its advantages are durable retention, repeated reads by independent consumers, and a mature ecosystem for analytics and downstream integration. The disadvantages are the extra operational footprint of a broker cluster, partition-local rather than global ordering, and a non-trivial tuning burden around acknowledgements, partitioning, batching, and replication. Recent review work also notes that Kafka benchmarks are frequently hard to reproduce because configuration details are underreported, which strengthens the case for a carefully controlled thesis benchmark [20]–[22].

C. RabbitMQ

Earlier comparative work positions RabbitMQ as a routing-oriented messaging system that offers flexible exchanges, acknowledgements, and strong support for enterprise-style asynchronous messaging [20], [23]. Current RabbitMQ documentation additionally introduces *streams* as an append-only, non-destructive, always persistent and replicated data structure that complements traditional queues rather than replacing them. Stream consumers can attach at arbitrary offsets or timestamps, replay data, and scale out via super streams when partitioning is needed [24].

This makes RabbitMQ attractive as a hybrid comparison point between classic broker semantics and newer log-like stream semantics within the same ecosystem. Its advantages are flexible routing, explicit acknowledgements, and a gradual migration path from queue-oriented messaging to append-only streams. The limitations are that classic RabbitMQ semantics are not optimised for replay-heavy logs, while stream mode introduces its own replication and partitioning trade-offs. RabbitMQ should therefore be evaluated not only on raw throughput, but also on how well it balances routing flexibility, operational continuity, and stream-specific behaviour [20], [23], [24].

D. NATS with JetStream persistence

The NATS ecosystem deserves separate consideration because it targets lightweight, location-transparent distributed communication. Official NATS documentation presents core NATS as a connective technology for modern distributed systems with publish/subscribe and request/reply communication, while JetStream adds persistence, replay, replication, and consumer state on top of the base message bus [25]–[27]. JetStream explicitly supports replay from the beginning of a stream, from a specific sequence number, or from a time-based position, and it offers both push-based and pull-based consumers for different workload profiles [27].

For the thesis, NATS is interesting because it represents a lightweight alternative to Kafka and RabbitMQ rather than a direct substitute for either. JetStream supports durable streams, replay, replication, and scalable pull consumers, while core NATS preserves a simple request/reply and pub/sub model that is attractive for microservice systems [26], [27]. A key complication is that comparative literature still often discusses the older NATS Streaming layer rather than JetStream, for example Sharvari and Sowmya Nag [23], which means that published results do not always map perfectly to the current product. NATS with JetStream is most promising when lightweight deployment, elastic scaling, and a smaller operational surface are valued more highly than Kafka’s larger ecosystem and benchmarking history.

E. Benchmark Design

1) *Methodological positioning and scope:* The benchmark design adapts recurring evaluation principles to the specific constraints of DYNAMOS rather than copying a single prior protocol. The controlled multi-stage pipeline follows the style of prior streaming-pipeline evaluations such as DataXc [7]; identical business logic across mechanisms follows comparative benchmarking practice for distributed systems, where Karimov et al. require the same workload definitions and measurement points across systems under test, and Henning and Hasselbring define scalability as the rate of throughput growth per added resource unit [28], [29]; the emphasis on throughput, latency, concurrency, resource pressure, and recovery reflects recurring evaluation dimensions in streaming and IPC (Inter-process communication) studies [17], [22].

The aim is not to identify a universally best mechanism, but to narrow the design space for DYNAMOS by asking which communication model best fits large, ordered, and concurrent inter-service transfer under realistic operational constraints. Five transport mappings are evaluated against the same pipeline, the same payload definitions, and the same correctness criteria, and the results are interpreted in terms of DYNAMOS integration. The chapter body keeps the main comparison compact: the completed single-node clean matrix, three focused operational scenarios, and the resulting architectural interpretation. Full preset definitions and detailed per-case tables remain in the appendix.

2) *Common testbed and fairness constraints:* All candidate mechanisms were evaluated in the same Go-based microservice chain deployed on Kubernetes: *Producer* → *Transformer* → *Sink*. The services ran in a dedicated namespace using the same manifests, the same payload schema, and the same transformation logic. Resource budgets were controlled through Kustomize overlays so that the comparison remained centered on the communication layer rather than on application-level differences [28], [29].

The transport mappings were also kept as close as possible to one another. Client-streaming and unary gRPC provided the direct, service-to-service reference points. Kafka preserved per-flow ordering through keyed partitioning. RabbitMQ used streams rather than classic queues. NATS used JetStream with explicit acknowledgement semantics. This alignment matters because it limits the risk that one mechanism benefits from a materially easier application mapping than the others.

The service logic remained intentionally lightweight. The producer emitted synthetic or replayed records, the transformer applied a deterministic operation, and the sink recorded latency, pacing, and correctness summaries. That kept the benchmark focused on transfer behaviour, buffering, and recovery rather than on domain-specific computation [7], [17].

3) *Evaluation dimensions:* The literature consistently treats streaming evaluation as a multi-dimensional problem rather than a single throughput race [7], [17], [22]. This study follows that structure. Five dimensions matter most here: transfer efficiency, latency and pacing stability, overload behaviour, recovery correctness, and operational cost. A transport that leads the clean matrix but degrades badly under backpressure or restart is not automatically the best fit for DYNAMOS. The benchmark therefore keeps those dimensions visible throughout the chapter.

4) *Scenario set and scenario-to-property mapping:* The completed RQ1 study uses four scenario families: a clean bulk-transfer matrix, a steady-rate continuous scenario, a slow-consumer backpressure scenario, and a forced-restart recovery scenario. Together they cover the properties most relevant to DYNAMOS: sustained transfer, regular pacing, overload handling, and behaviour during interruption.

5) *Experimental factors:* The main independent variables are payload size, concurrency, target rate where applicable, sink slowdown in the backpressure case, and explicit failure injection in the recovery case. Resource budgets are held

Table I
SCENARIO-TO-PROPERTY MAPPING USED IN THE RQ1 BENCHMARK.

Scenario	Primary property	Why it matters for DYNAMOS
Clean bulk transfer	Protocol overhead, sustained throughput, multi-hop transfer efficiency	Large dataset exchange should remain efficient when data moves incrementally through a composed service chain.
Continuous steady-rate flow	Latency stability, inter-arrival regularity, pacing precision	Long-lived flows matter when downstream services depend on a predictable delivery rhythm.
Slow-consumer backpressure	Queue build-up, overload propagation, throttling versus delay accumulation	DYNAMOS pipelines can contain stages with unequal service rates, so overload handling must remain interpretable.
Forced-restart recovery	Reconnect behaviour, duplicates, ordering preservation, recovery time	Ephemeral microservices and restarts are normal in Kubernetes-style deployments.

constant through Kubernetes overlays so that performance differences reflect the transport mapping rather than an uneven infrastructure allocation. This keeps the experiment matrix broad enough to expose real tradeoffs without turning the chapter into an exhaustive lab log [22], [23].

6) *Result presentation and appendix boundary:* The main text reports one clean-summary table, one compact size-wise clean breakdown, and three focused scenario tables for continuous flow, backpressure, and recovery. Detailed per-case clean-matrix tables remain in Appendix B. Broker resource figures are not used in the main argument because the currently available exported broker CPU series are not yet semantically comparable across transports.

F. Results

1) *Metric definitions:* The core metrics are throughput in messages per second and MiB per second, end-to-end latency percentiles, inter-arrival regularity, duplicate count, and ordering violations. Throughput indicates transfer efficiency. Latency percentiles show how quickly messages move through the pipeline. Inter-arrival and jitter show how regular that delivery remains over time. Duplicates and ordering violations capture correctness. The recovery scenario additionally separates pre-failure, recovery-window, and post-recovery throughput; time to first post-failure message; time to a sustained five-second window at or above 80% of the target rate; throughput debt inside the fixed recovery window; recovery-window tail latency; post-failure correctness counts; backlog drain time; and completion.

2) *Comparison scope:* The primary comparison in this chapter is the completed single-node Kubernetes benchmark across five transports: client-streaming gRPC, unary gRPC, RabbitMQ Streams, NATS JetStream, and Kafka. The three diagnostic scenario tables (continuous, backpressure, recovery) report the latest single-node run executed on 4 May 2026, with three repetitions per transport. The clean-matrix summary in Table II and the per-size breakdown in Table III, together with the corresponding appendix tables, retain the earlier completed multi-cell run because not all broker cells were re-executed in the most recent refresh; the qualitative ranking on the clean matrix is consistent across the two runs, but absolute throughput levels differ and the reader should not directly

cross-compare absolute numbers between the headline clean-matrix tables and the diagnostic scenario tables. Exploratory clustered broker runs were also executed; they are not part of the primary results because coverage was limited and startup or convergence effects materially influenced some first runs, and they are discussed briefly in the interpretation section rather than presented as headline evidence.

Table II shows a clear separation between the two direct gRPC mappings and the three brokered alternatives. Client-streaming is the fastest transport in the clean matrix by a wide margin. Unary is substantially slower, but it does so with very low tail latency. Among the brokered transports, NATS JetStream achieves the highest average throughput, followed by RabbitMQ Streams and Kafka. Ordering violations are absent in the clean matrix. The remaining correctness anomalies are concentrated in duplicate deliveries on NATS and, to a much smaller extent, Kafka.

3) *Clean bulk-transfer matrix:* The size-wise breakdown in Table III shows how the clean-matrix ranking changes with message size. Client-streaming remains the fastest option at every size. Its advantage over unary is largest at small payloads, where per-request overhead dominates, and narrows as the payload grows. NATS JetStream is the strongest brokered transport across all four size bands. RabbitMQ Streams is competitive at 256 B and 1 KiB, then declines more sharply at larger payloads. Kafka stays below NATS in every size tier, but remains broadly stable and avoids the sharp collapse seen in RabbitMQ at 16 KiB.

All clean-matrix runs reached the target message count. Full per-case tables remain in Appendix B, but the main pattern is already visible here: client-streaming dominates bulk transfer, unary remains the low-latency reference rather than a throughput candidate, and the three brokered transports form a separate performance band with NATS at its front.

4) *Continuous steady-rate scenario:* The continuous scenario uses 1 KiB messages and a target rate of 1,000 msg/s. The purpose is not to maximize throughput, but to see how steadily each transport sustains an ongoing flow.

All five transports sustain the 1,000 msg/s target and overshoot it slightly (1,033–1,053 msg/s) in this run, so the comparison is dominated by latency stability rather than by who can keep up. Client-streaming and unary have by far the lowest

Table II
SINGLE-NODE SYNTHETIC-CLEAN SUMMARY, AVERAGED ACROSS ALL PAYLOAD SIZES AND CONCURRENCY LEVELS.

Transport mode	Avg. msg/s	Avg. MiB/s	Avg. p_{95}	Avg. p_{99}	Duplicates	Ordering viol.
Client-streaming	31,259.76	51.35	242.82	441.59	0	0
Unary	2,291.37	9.55	3.01	31.23	0	0
RabbitMQ Streams	9,110.73	16.29	2,123.56	2,303.05	0	0
NATS JetStream	13,116.27	32.13	1,581.11	1,659.99	5,086	0
Kafka	7,354.42	20.12	1,729.34	1,901.42	9	0

Table III
SINGLE-NODE SYNTHETIC BULK-TRANSFER THROUGHPUT BY MESSAGE SIZE, AVERAGED ACROSS CONCURRENCY LEVELS 1, 4, 8, AND 16.

Msg size	Stream msg/s	Unary msg/s	RabbitMQ msg/s	NATS msg/s	Kafka msg/s
256 B	75,604	2,586	13,065	22,561	12,467
1 KiB	27,682	2,599	14,381	15,432	7,573
4 KiB	15,358	2,319	7,905	10,086	6,527
16 KiB	6,395	1,662	1,092	4,387	2,850

Table IV
CONTINUOUS STEADY-RATE SCENARIO: SINGLE-NODE RESULTS (1 KiB PAYLOAD, 1,000 MSG/S TARGET, ONE REPETITION PER CELL). THROUGHPUT IN MSG/S; LATENCY, INTER-ARRIVAL, AND JITTER VALUES IN MILLISECONDS.

Transport mode	Concur.	Throughput	Avg. p_{95}	p_{95} inter-arrival	Jitter
Client-streaming	4	1,049.69	7.16	1.66	0.68
Client-streaming	8	1,050.93	7.14	1.65	0.70
Unary	4	1,050.84	0.88	2.10	1.09
Unary	8	1,050.73	0.91	2.11	0.99
RabbitMQ Streams	4	1,047.83	34.07	2.89	1.30
RabbitMQ Streams	8	1,049.16	49.36	3.51	2.11
NATS JetStream	4	1,052.71	26.67	1.95	0.84
NATS JetStream	8	1,052.03	102.25	3.07	2.71
Kafka	4	1,042.55	326.52	0.01	8.22
Kafka	8	1,033.59	387.36	0.03	7.94

tail latency, with unary p_{95} below 1 ms and client-streaming p_{95} around 7 ms in both concurrency settings. Among the brokered transports, NATS at concurrency 4 (26.7 ms p_{95}) and RabbitMQ Streams (34–49 ms p_{95}) are close to one another, while NATS at concurrency 8 widens to 102 ms. Kafka remains the slowest in the tail, with p_{95} between 327 ms and 387 ms; its inter-arrival distribution is dominated by short bursts (0.01–0.03 ms p_{95}) and its jitter is the highest of any transport (around 8 ms), reflecting Kafka’s batched delivery pattern.

5) *Backpressure scenario*: The backpressure scenario keeps the payload at 1 KiB and concurrency at 8, but adds a 2 ms sink delay while the producer targets 4,000 msg/s. This exposes how each transport behaves once downstream processing becomes the bottleneck.

Table V
BACKPRESSURE SCENARIO: SINGLE-NODE RESULTS (1 KiB, CONCURRENCY 8, 4,000 MSG/S TARGET, 2 MS SINK DELAY, ONE REPETITION PER TRANSPORT). THROUGHPUT IN MSG/S; LATENCY, INTER-ARRIVAL, AND JITTER VALUES IN MILLISECONDS.

Transport mode	Throughput	Median lat.	Avg. p_{95}	p_{95} inter-arrival	Jitter
Client-streaming	3,333.93	1,443.26	1,878.90	2.14	0.64
Unary	2,091.96	0.53	1.41	3.40	1.23
RabbitMQ Streams	2,557.86	2,603.70	3,476.80	2.26	1.68
NATS JetStream	2,973.05	1,768.99	3,034.92	2.14	0.69
Kafka	1,268.34	4,879.51	11,363.59	2.53	1.32

Client-streaming, NATS, and RabbitMQ Streams all keep their realised throughput in the 2.5–3.3 k msg/s range under pressure, well below the 4,000 msg/s target but clearly above the unary baseline. They achieve this by absorbing queueing delay: medians around 1.4–2.6 s and p_{95} values between 1.9 s and 3.5 s. Unary gives up throughput earlier (2,092 msg/s) and therefore stays extremely low-latency for the requests that complete (median 0.5 ms, p_{95} 1.4 ms), which is the expected effect of per-request acknowledgement under sink slowdown. Kafka is the most throughput-constrained transport in this single-node backpressure case (1,268 msg/s) and develops the heaviest tail of any transport (p_{95} above 11 s), again reflecting its batched delivery pattern under load.

6) *Recovery scenario*: The recovery scenario uses 4 KiB messages at 2,000 msg/s and forces a transformer restart after 3 seconds. The table separates the run into pre-failure throughput, a fixed 10 s recovery window, and the post-recovery tail so that restart visibility, sustained recovery, backlog drain, and correctness cost are not collapsed into one ambiguous number.

Before failure, all transports are close enough to the configured 2,000 msg/s target to make the recovery comparison meaningful. The disruption then separates the mechanisms sharply. Client-streaming produces the fastest visible resumption, with the first post-failure message arriving within 1 ms, but its recovery-window throughput falls to 939 msg/s and it accumulates 10,630 messages of throughput debt. Its 10,615 post-failure duplicates are a direct consequence of the current reconnect policy: after a stream send fails, the producer reopens the stream and replays the worker stream rather than resuming exactly at the failed chunk. The historical 20,000-message run ends before five complete stable buckets can

Table VI

RECOVERY SCENARIO: SINGLE-NODE RESULTS (4 KiB, CONCURRENCY 8, 2,000 MSG/S, TRANSFORMER RESTART AT 3 s). ONE REPETITION PER TRANSPORT. THROUGHPUT VALUES ARE IN MSG/S; TIME, LATENCY, AND DRAIN VALUES ARE IN MILLISECONDS. SUSTAINED RECOVERY IS THE FIRST FIVE COMPLETE ONE-SECOND BUCKETS INSIDE THE CONFIGURED TARGET BAND, WITH NO FALLBACK TO BACKLOG DRAIN. DASHES INDICATE THAT THE VALUE WAS NOT OBSERVED IN THE RUN OR THAT NO MESSAGES ARRIVED IN THAT PHASE/WINDOW.

Transport mode	Pre msg/s	Recovery msg/s	Post msg/s	First post-fail	Sustained 80%/5s	Debt	Recovery p_{95}	Post-fail dup.	Post-fail order	Backlog drain	Done
Client-streaming	2,711	939	1,999	0.18	–	10,630	2.74	10,615	0	11,446.21	Yes
Unary	2,161	0	777	21,824.57	22,000	20,000	–	0	0	27,734.03	Yes
RabbitMQ Streams	1,778	2,011	–	292.74	–	4,868	3,921.74	886	0	7,463.72	Yes
NATS JetStream	2,297	1,351	0.10	2.56	–	7,847	388.89	0	2	29,228.03	Yes
Kafka	2,343	0	710	29,224.26	–	20,000	–	62	0	29,760.29	Yes

be observed after the later recovery point, so the sustained-recovery field is intentionally blank rather than inferred.

Unary shows the opposite shape. It preserves correctness, but it delivers no messages in the fixed recovery window, waits roughly 21.8 s before the first post-failure message, and accumulates the full 20,000-message recovery-window debt. It eventually records a sustained five-second window at 22 s, but by then the useful distinction is already visible: unary turns the restart into an extended outage rather than a buffered drain.

The brokered transports expose three different buffering behaviours. RabbitMQ Streams resumes within 293 ms and drains the run fastest, with recovery-window throughput slightly above target, but it pays for that with 886 post-failure duplicates and a recovery-window p_{95} latency near 3.9 s. NATS JetStream is the case where the new metrics matter most: the first post-failure message arrives after only 2.6 ms, but the fixed recovery window still shows 7,847 messages of debt and no sustained recovery because the early replay plateau is followed by an extended quiet period. Kafka is the most delayed in this run: no messages arrive during the recovery window, the first post-failure message appears after about 29.2 s, and the run remains backlog-dominated until completion, although it keeps post-failure duplicates low and preserves ordering.

7) *Discussion:* The benchmark and the DYNAMOS architecture do not point to exactly the same answer, and that distinction matters. In the isolated pipeline benchmark, client-streaming gRPC is the strongest raw transport: it leads the clean matrix, remains stable in the continuous scenario, and avoids the broker overheads seen in the other transports. If RQ1 were answered only at the level of transport efficiency in a controlled pipeline, client-streaming would be the clearest winner.

DYNAMOS as implemented today, however, has two different communication regimes. Within a deployed microservice-chain job, services already communicate locally over gRPC inside a single Pod, where latency is negligible and where chunked gRPC transfer is a natural fit. Across agents, by contrast, DYNAMOS currently relies on sidecar-mediated RabbitMQ communication, ephemeral per-job queue management, and retry logic that explicitly tolerates the fact that the

destination job may not yet be ready when data is sent. The main large-object transfer problem in current DYNAMOS is therefore not an already-local in-Pod hop, but the inter-agent exchange of bounded SQL-style result sets that are currently packed into a single protobuf payload and forwarded through the brokered path.

Viewed through that architectural lens, RabbitMQ Streams becomes the strongest first implementation candidate for DYNAMOS, even though it does not win the benchmark outright. It preserves temporal decoupling between producer and consumer, fits naturally with ephemeral jobs and cold-start behaviour, reuses the current sidecar and RabbitMQ operational model, and still enables incremental chunked transfer so that peak memory usage can be reduced. It also offers a more natural path toward auditability and replay than direct RPC if future compliance requirements require message-level evidence or reconstruction of inter-agent transfers. Its downside is lower raw transport efficiency than client-streaming and the continued presence of broker latency and broker resource cost.

Client-streaming gRPC nevertheless remains a serious option, but as a more targeted optimisation than as an unconditional system-wide recommendation. Its main attraction for DYNAMOS is efficient chunked transfer with immediate pipelining once both endpoints are alive. Its main cost is that DYNAMOS would need to re-implement, at application level, some of the temporal decoupling that RabbitMQ already provides: reconnect handling, replay boundaries, and idempotent recovery when the destination job is not yet ready or restarts mid-transfer. The recovery benchmark already shows that these concerns are not theoretical. For DYNAMOS, client-streaming therefore looks strongest as a second MVP to compare against RabbitMQ Streams on the specific inter-agent large-result path, rather than as the obvious first transport to standardise across the system.

Kafka and NATS JetStream are lower-priority options for the current DYNAMOS architecture. Both would require a more substantial platform change, and both bring strengths, large-scale replay, log-centric consumption, multi-consumer stream processing, or subject-based pub/sub flexibility, that DYNAMOS' present request-shaped SQL workloads do not

strongly exploit. They remain valuable comparison points in the benchmark and possible future directions if DYNAMOS evolves toward persistent, replay-heavy, or multi-consumer workloads, but they do not appear to be the most pragmatic next step for the system as it exists today.

The most defensible next step is therefore a staged RQ2 integration strategy: preserve classic RabbitMQ queues for the control plane, prototype RabbitMQ Streams for the inter-agent data path as the lowest-risk architectural evolution, and build a smaller client-streaming gRPC MVP for the same path as a targeted alternative. RQ3 can then evaluate both end-to-end inside DYNAMOS, where startup behaviour, policy enforcement, failure handling, and real workload structure can be weighed against the isolated transport-level results reported here.

G. Threats to Validity

Three limits are important when interpreting these results. First, the primary five-transport comparison is based on completed single-node runs, and the focused operational scenarios are point measurements rather than confidence-interval estimates. They are strong enough for comparative interpretation, but not for a fully statistical treatment. Second, the benchmark pipeline is intentionally simple. That makes transport behaviour easier to isolate, but it does not capture every co-ordination cost that appears in a full DYNAMOS deployment. Third, the cross-transport *ranking* of broker CPU usage in the clean matrix is not used as headline evidence, because the currently exported CPU series are not yet semantically aligned across transports at that scale. The resource figures cited from Appendix D in the operational-cost argument are restricted to the diagnostic continuous and recovery scenarios and are used only to indicate the order of magnitude of broker overhead, not to compare brokers against each other on CPU efficiency.

IV. CONCLUSION

RQ1 asks which streaming communication technologies, frameworks, and interaction patterns are suitable for large-scale and continuous data exchange in microservice-based systems such as DYNAMOS. At the level of isolated transport performance, client-streaming gRPC is the strongest result in this chapter: it provides the highest sustained throughput on the clean matrix and the most stable pacing in the continuous scenario. That makes it the clearest benchmark winner.

For DYNAMOS as it is currently implemented, however, the most plausible first integration candidate is RabbitMQ Streams rather than client-streaming gRPC. DYNAMOS already depends on broker-mediated inter-agent communication, ephemeral jobs, sidecar abstraction, and queue or routing-key based decoupling between sender and receiver. RabbitMQ Streams preserves these properties while still enabling chunked, memory-efficient transfer of large bounded result sets. Client-streaming gRPC remains an attractive targeted alternative, especially for large inter-agent transfers where pipelining matters more than decoupling, but it would require more application-level recovery logic to fit cleanly into DYNAMOS’ existing

operational model. Kafka and NATS JetStream remain useful benchmark references and possible longer-term directions, but they appear to require more architectural change than their current DYNAMOS-specific benefit justifies.

The recommendation made here is therefore deliberately two-level and conditional. The benchmark winner is client-streaming gRPC; the current system-fit recommendation for DYNAMOS is to prototype RabbitMQ Streams first for the inter-agent data path, keep classic RabbitMQ queues for the control plane, and build a smaller client-streaming MVP as a direct comparison before committing to one design. The remainder of the thesis addresses that decision process. RQ2 turns these candidate communication models into concrete integration designs for DYNAMOS, with attention to dynamic service composition, policy enforcement, recovery, and life-cycle management. RQ3 then evaluates them at full system level, asking which option offers the best overall trade-off once orchestration overhead, middleware behaviour, multi-node deployment, and real DYNAMOS workloads are included. The architectural choice is therefore not yet a closed result; it is a benchmark-informed hypothesis that must be validated inside DYNAMOS itself.

APPENDIX A BENCHMARK PRESET DEFINITIONS

This appendix documents the complete preset definitions used in the benchmark. Each preset specifies the workload family, payload configuration, concurrency level, target rate (where applicable), failure injection parameters, and resource overlay tier. The five presets are: *synthetic-clean* (bulk transfer baseline), *csv-replay-check* (realism validation), *synthetic-continuous* (steady-rate pacing), *synthetic-backpressure* (slow-consumer overload), and *synthetic-recovery* (forced-restart resilience).

Table VII
BENCHMARK PRESETS USED IN THE CURRENT TRANSPORT STUDY.

Preset	Key parameters	Purpose
<i>synthetic-clean</i>	Synthetic payloads from 256 B to 16 KiB, concurrency 1–16, no injected failure	Baseline bulk-transfer throughput, latency, and correctness matrix
<i>csv-replay-check</i>	Copied DYNAMOS CSV rows batched as 25 or 100 rows per message	Realism check against non-synthetic payload structure
<i>synthetic-continuous</i>	1 KiB payloads, 1,000 msg/s target, concurrency 4 and 8, one completed comparable run	Steady-rate pacing and jitter comparison
<i>synthetic-backpressure</i>	1 KiB payloads, 4,000 msg/s target, concurrency 8, 2 ms sink delay, one completed comparable run	Slow-consumer overload and queueing behaviour
<i>synthetic-recovery</i>	4 KiB payloads, 2,000 msg/s target, concurrency 8, transformer restart after 3 s, 120 retries at 250 ms backoff, one completed comparable run	Restart recovery, replay, and continuity analysis

APPENDIX B FULL RUN MATRIX

This appendix summarizes the canonical run scopes used in the current thesis. The body of the thesis presents the most

decision-relevant subsets, while the benchmark result directory stores the corresponding per-run JSON, NDJSON, and CSV exports generated by the execution scripts. The chapter is based on the single-node five-transport `synthetic-clean` matrix, the single-node diagnostic scenario suite, and the three-broker clustered scenario runs with three repetitions per transport per scenario.

Table VIII
EXECUTION SCOPE OF THE CURRENT BENCHMARK CORPUS.

Scope	Transports	Variable dimensions	Reps	Role in thesis
Single-node clean matrix	Client-streaming, unary, RabbitMQ, NATS, Kafka	Payload 256 B–16 KiB, concurrency 1–16	1	Full bulk-transfer baseline (Section III)
Single-node CSV replay	All five	Batch size 25 and 100, concurrency subset	1	Realism validation
Single-node continuous	All five	Concurrency 4 and 8, 1 KiB, 1,000 msg/s	1	Pacing and jitter evidence
Single-node backpressure	All five	1 KiB, concurrency 8, 4,000 msg/s, 2 ms sink delay	1	Overload and queueing evidence
Single-node recovery	All five	4 KiB, concurrency 8, transformer restart	1	Recovery and correctness evidence
Clustered clean (3-broker)	RabbitMQ, NATS, Kafka	4 KiB, concurrency 8	3	Clustered cross-run bulk-transfer consistency (Section III)
Clustered continuous (3-broker)	RabbitMQ, NATS, Kafka	1 KiB, concurrency 8, 1,000 msg/s	3	Clustered pacing under broker replication
Clustered backpressure (3-broker)	RabbitMQ, NATS, Kafka	1 KiB, concurrency 8, 4,000 msg/s, 2 ms delay	3	Clustered overload behaviour
Clustered recovery (3-broker)	RabbitMQ, NATS, Kafka	4 KiB, concurrency 8, transformer restart	3	Clustered recovery correctness and variability

APPENDIX C DETAILED CLEAN-MATRIX RESULTS

This appendix records the full completed five-transport `synthetic-clean` matrix at the per-case level. The main chapter uses averages by transport and message size to keep the argument readable, but the tables below expose the exact throughput and latency values for every payload-size and concurrency combination in the current single-repeat suite.

Table IX
DETAILED SYNTHETIC-CLEAN MESSAGE THROUGHPUT (MSG/S) BY PAYLOAD SIZE AND CONCURRENCY.

Msg size	Concur.	Stream	Unary	RabbitMQ	NATS	Kafka
256 B	1	76,468	637	3,058	19,548	15,768
	4	118,209	1,899	25,812	24,292	10,876
	8	66,768	3,530	21,838	5,757	12,565
	16	40,970	4,279	1,553	28,496	10,660
1 KiB	1	28,171	696	17,810	16,976	10,760
	4	36,448	2,161	23,577	15,253	9,688
	8	28,743	3,163	1,045	6,350	5,103
	16	17,368	4,375	15,091	3,601	4,740
4 KiB	1	16,424	673	5,975	11,101	6,377
	4	19,838	1,834	8,928	11,211	6,903
	8	15,930	2,945	8,859	12,002	7,865
	16	9,239	3,823	7,860	5,360	4,965
16 KiB	1	2,872	635	2,233	4,497	2,737
	4	9,487	1,570	769	3,866	2,866
	8	9,138	1,698	466	6,507	2,797
	16	4,083	2,745	899	5,257	3,000

Table X
DETAILED SYNTHETIC-CLEAN DATA THROUGHPUT (MiB/S) BY PAYLOAD SIZE AND CONCURRENCY.

Msg size	Concur.	Stream	Unary	RabbitMQ	NATS	Kafka
256 B	1	18.67	0.16	0.75	4.77	3.85
	4	28.86	0.46	6.30	5.93	2.66
	8	16.30	0.86	5.33	1.41	3.07
	16	10.00	1.04	0.38	6.96	2.60
1 KiB	1	27.51	0.68	17.39	16.58	10.51
	4	35.59	2.11	23.02	14.90	9.46
	8	28.07	3.09	1.02	6.20	4.98
	16	16.96	4.27	14.74	3.52	4.63
4 KiB	1	64.16	2.63	23.34	43.36	24.91
	4	77.49	7.16	34.88	43.79	26.97
	8	62.23	11.50	34.61	46.88	30.72
	16	36.09	14.93	30.70	20.94	19.39
16 KiB	1	44.88	9.92	34.89	70.27	42.76
	4	148.24	24.54	12.01	60.41	44.78
	8	142.78	26.53	7.28	101.67	43.71
	16	63.79	42.89	14.05	82.14	46.88

Table XI
DETAILED SYNTHETIC-CLEAN p_{95} LATENCY (MS) BY PAYLOAD SIZE AND CONCURRENCY.

Msg size	Concur.	Stream	Unary	RabbitMQ	NATS	Kafka
256 B	1	44.4	1.9	1,699.4	602.1	205.8
	4	63.7	3.2	481.8	457.1	1,112.3
	8	97.6	2.2	590.8	3,296.7	999.7
	16	227.1	3.9	1,111.3	422.6	941.1
1 KiB	1	31.6	1.8	314.2	661.1	476.4
	4	183.9	1.6	496.3	955.7	797.5
	8	185.0	2.9	1,833.1	2,773.8	3,511.2
	16	681.1	4.5	786.6	5,249.4	3,607.1
4 KiB	1	23.3	1.7	1,526.0	979.1	768.8
	4	66.0	3.1	998.3	1,160.2	1,438.7
	8	333.7	2.4	1,173.3	1,023.2	1,304.8
	16	1,067.7	4.1	1,589.8	3,256.2	3,039.3
16 KiB	1	35.0	1.5	1,001.1	2,226.7	138.5
	4	153.8	3.3	13,731.6	3,890.3	3,515.6
	8	300.5	4.7	407.2	1,933.2	2,789.9
	16	390.7	5.5	6,236.1	2,448.1	3,022.5

APPENDIX D RESOURCE UTILISATION DETAILS

This appendix summarizes resource metrics for the diagnostic Kubernetes scenario suite where CPU and memory data are most informative. For brokered modes, the broker is reported as a separate role because its buffering layer is a core part of the transport cost. The backpressure preset produced too few stable `kubect1 top` samples to support a meaningful per-role comparison across all transports, so the appendix focuses on the currently available continuous and recovery scenarios across all five transports. A clean-matrix

Table XII
DETAILED SYNTHETIC-CLEAN p_{99} LATENCY (MS) BY PAYLOAD SIZE AND CONCURRENCY.

Msg size	Concur.	Stream	Unary	RabbitMQ	NATS	Kafka
256 B	1	77.7	12.2	1,700.3	616.4	311.0
	4	65.8	15.5	507.5	501.6	1,210.1
	8	101.0	27.3	622.2	3,320.0	1,115.3
	16	266.6	48.5	1,123.1	487.0	1,097.6
1 KiB	1	57.0	13.1	338.3	723.4	484.5
	4	190.2	4.8	534.5	965.1	814.8
	8	249.8	41.2	1,909.0	2,865.2	3,624.5
	16	745.8	36.1	812.4	5,302.7	4,009.9
4 KiB	1	39.0	14.7	1,573.9	1,015.6	791.9
	4	76.3	23.9	1,191.1	1,182.2	1,516.1
	8	399.6	53.7	1,220.7	1,103.0	1,591.3
	16	1,379.0	64.8	1,934.5	3,348.0	3,167.0
16 KiB	1	44.8	15.4	1,041.4	2,361.8	294.2
	4	188.2	28.7	14,750.1	3,952.5	3,800.5
	8	347.6	33.9	598.8	2,012.5	3,173.7
	16	2,837.1	65.9	6,991.1	2,782.5	3,420.5

broker CPU summary is intentionally omitted because the currently exported broker CPU series are not semantically comparable enough to support a defensible cross-transport claim.

Table XIII
CONTINUOUS SCENARIO CPU USAGE BY ROLE (AVG./PEAK %).

Mode	Concur.	Producer	Transformer	Sink	Broker
Client-streaming	4	23.40 / 23.40	8.99 / 24.20	5.61 / 14.90	–
Client-streaming	8	–	9.31 / 25.60	5.64 / 15.50	–
Unary	4	35.70 / 35.70	17.88 / 39.10	11.12 / 24.10	–
Unary	8	–	15.54 / 33.70	9.72 / 20.90	–
RabbitMQ Streams	4	7.92 / 8.30	9.87 / 15.60	6.91 / 10.90	29.88 / 43.30
RabbitMQ Streams	8	7.88 / 8.20	10.46 / 15.80	7.50 / 11.70	33.00 / 46.50
NATS JetStream	4	–	12.62 / 24.90	5.60 / 15.40	18.37 / 40.90
NATS JetStream	8	–	16.23 / 23.70	6.45 / 14.20	22.74 / 38.50
Kafka	4	–	4.68 / 5.70	0.69 / 1.90	51.52 / 52.90
Kafka	8	–	8.58 / 8.80	1.00 / 2.20	41.13 / 58.80

Table XIV
RECOVERY SCENARIO CPU USAGE BY ROLE (AVG./PEAK %).

Mode	Producer	Transformer	Sink	Broker
Client-streaming	31.60 / 31.60	12.64 / 31.60	3.89 / 17.50	–
Unary	4.60 / 6.00	2.45 / 9.10	2.56 / 6.10	–
RabbitMQ Streams	8.84 / 10.20	13.38 / 17.10	7.51 / 13.90	31.76 / 49.90
NATS JetStream	0.30 / 0.30	17.29 / 25.20	7.28 / 16.20	20.99 / 43.50
Kafka	0.40 / 0.40	7.63 / 8.10	1.44 / 2.50	46.77 / 75.40

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Table XV
RECOVERY SCENARIO MEMORY USAGE BY ROLE (AVG./PEAK MiB).

Mode	Producer	Transformer	Sink	Broker
Client-streaming	7.00 / 7.00	3.40 / 7.00	2.33 / 7.00	–
Unary	7.00 / 7.00	3.08 / 7.00	4.53 / 7.00	–
RabbitMQ Streams	10.00 / 10.00	25.72 / 31.00	5.78 / 8.00	490.91 / 511.00
NATS JetStream	8.00 / 8.00	14.00 / 14.00	9.75 / 15.00	306.06 / 537.00
Kafka	10.00 / 10.00	8.00 / 8.00	6.53 / 9.00	449.71 / 478.00

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